Computational Physics: Problem Solving with Julia

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Preface

This is a Quarto book.

To learn more about Quarto books visit https://quarto.org/docs/books.

1 Introduction

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

2 Differential Equations and Nonlinear Oscillations

Numerical methods for solving ordinary differential equations. Applying those methods to nonlinear systems.

```
using CairoMakie, LaTeXStrings, LinearAlgebra
```

2.1 Nonlinear Oscillations

$$F_k(x) + F_{\text{ext}}(x,t) = m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} \tag{2.1}$$

2.2 Orbit Around a Point Mass

```
const G = 1.0
const M = 1.0;
```

```
function f(t, state)
    pos, vel = state
    acc = - G * M * pos / norm(pos)^3
    return [vel, acc]
end;
```

```
mutable struct Star{T}
    state::Array{T}
    pos_hist
end;
```

```
function rk4(t, y, force, dt)
    """
    Function to calculate the next timestep using:
        4th-order Runge-Kutta
    """
    k1 = force(t, y)
    k2 = force(t + 0.5 * dt, y + 0.5 * k1 * dt)
    k3 = force(t + 0.5 * dt, y + 0.5 * k2 * dt)
    k4 = force(t + dt, y + k3 * dt)

    y_new = y + dt * (k1 + 2.0*k2 + 2.0*k3 + k4) / 6.0
end;
```

```
function orbit(t_start, t_end, dt, star, force)
    t = t_start
    while t < t_end
        new_state = rk4(t, star.state, force, dt)
        star.state = new_state
        push!(star.pos_hist, star.state[1])
        t += dt
    end
    star.pos_hist = reduce(hcat, star.pos_hist)'
end;
```

```
start_pos = [1.0, 0.0, 0.0]
start_vel = [0.0, 1.0, 0.0]
start_state = [start_pos, start_vel]
star = Star(start_state, [start_state[1]])

h = 1e-4
norbits = 1.0
t_end = 2 * pi * norbits
orbit(0.0, t_end, h, star, f);
```

```
function plot_orbit(star, name)
    inch = 96
    pt = 4/3

    fig = Figure(size=(10inch, 7inch), fontsize=16pt)
    ax = Axis(fig[1, 1],
            xlabel = "x",
```

```
            ylabel = "y",
            title=L"Orbit around a Point Mass ($\Delta t$ = %$(h))",
            aspect = DataAspect())

    plot!(ax, star.pos_hist[:, 1], star.pos_hist[:, 2])
    save(name, fig)
end;
```

```
with_theme(theme_latexfonts()) do
    plot_orbit(star, "images/rk4.svg")
end;
```

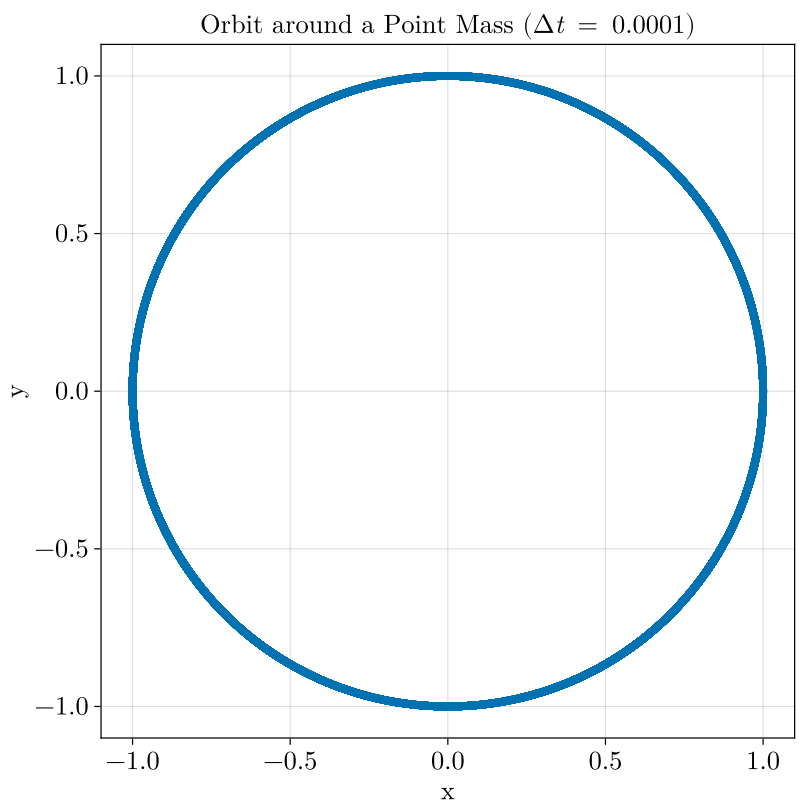

Figure 2.1: Orbit around a point mass using RK4

3 Summary

In summary, this book has no content whatsoever.

References

Knuth, Donald E. 1984. "Literate Programming." *Comput. J.* 27 (2): 97–111. https://doi.org/10.1093/comjnl/27.2.97.